

OTOMASI SISTEM PEMROSESAN DATA IKLIM MENGUNAKAN METODE SMOOTHING (STUDI KASUS: KABUPATEN ACEH BESAR)

Sidang Skripsi

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Pembimbing

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Outline

1. PERBAIKAN SEMINAR HASIL
2. HASIL DAN PEMBAHASAN
3. KESIMPULAN



Intisari Penelitian (Abstrak)

- Masalah: Fluktuasi padi Aceh Besar & data iklim belum siap pakai.
- Data: BMKG (7.565 baris) & NASA POWER (7.578 baris) 2005-2025
- Metode: Cubic Spline, Two-Stage Binary, Adaptive Exponential.
- Hasil: Suhu & Kelembaban BMKG >99% (excellent), NASA >82% (trend preserved).
- Output: Data siap peramalan & Kalender Tanam Digital.



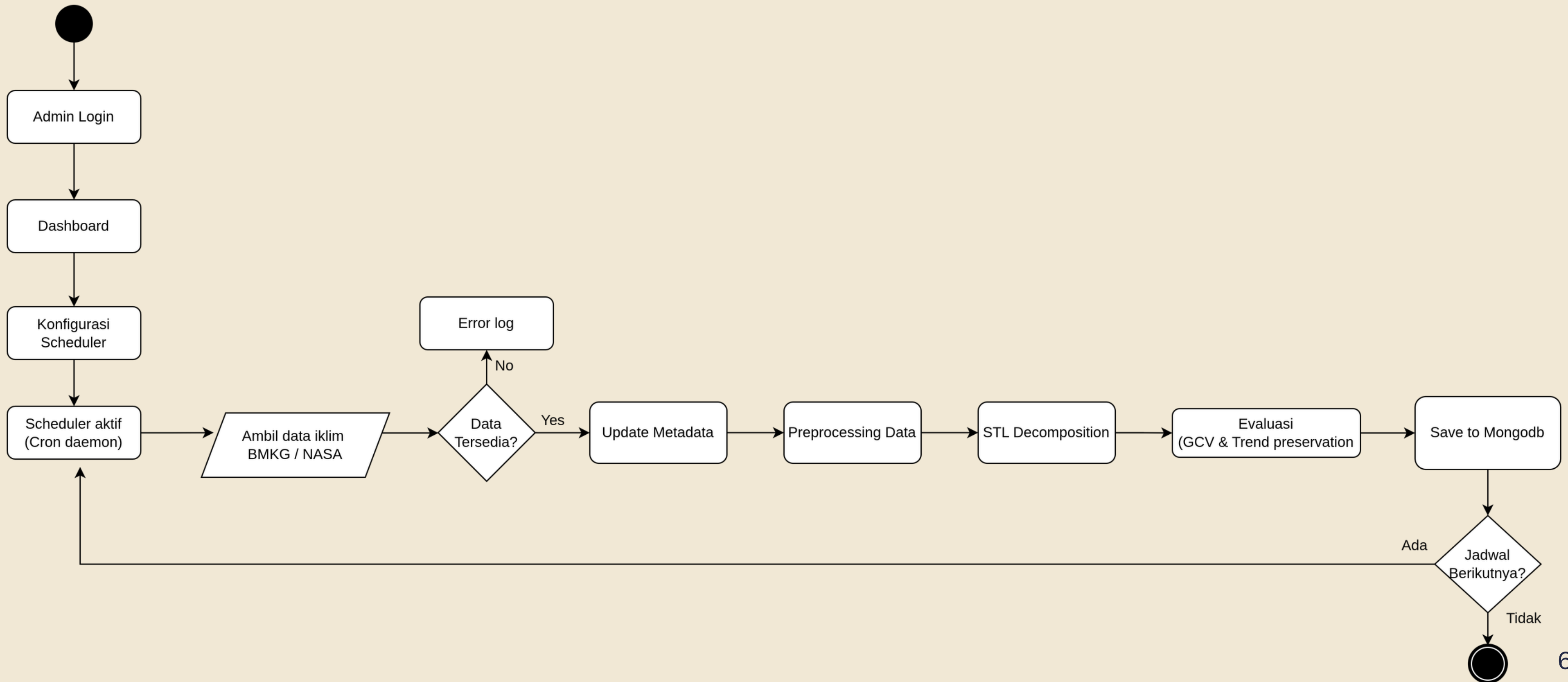
1. PERBAIKAN



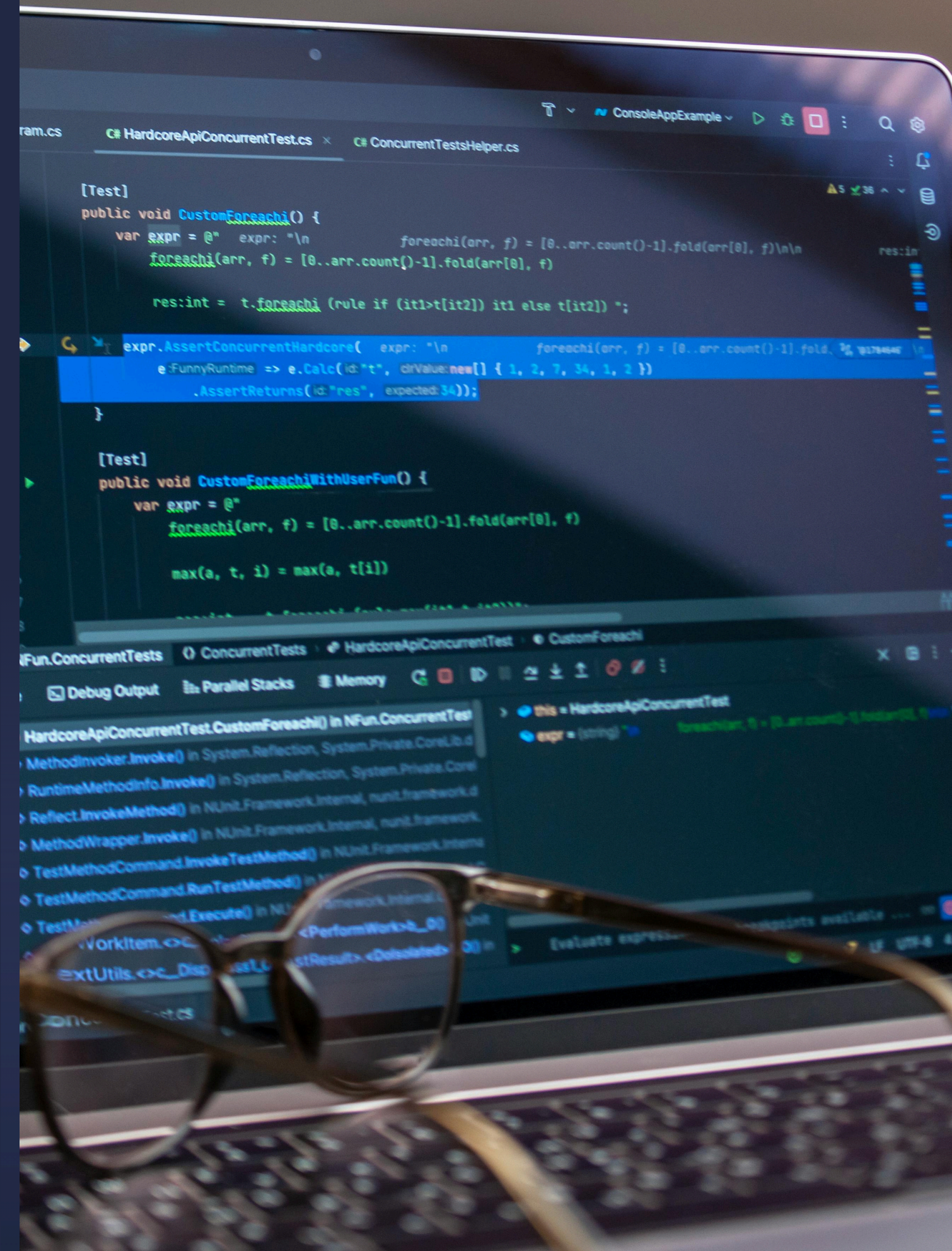
Perbaikan Seminar Hasil

No	Aspek Revisi	Tindak Lanjut Perbaikan
1	Flowchart Anggota 1	Menambahkan alur dekomposisi STL, memperjelas scheduler sebagai trigger utama, dan menambahkan loop "Jadwal Berikutnya".
2	Metode Smoothing - BAB II	Mengganti sub bab Single Exponential/HW/LSTM menjadi Cubic Spline + GCV (untuk BMKG) dan Adaptive Exponential Smoothing (untuk NASA).
3	Survei Kuesioner - Bab III	Menegaskan bahwa hasil kuesioner tidak digunakan untuk preprocessing data iklim (BMKG/NASA), melainkan untuk penyusunan kalender tanam digital dan identifikasi kebutuhan pengguna.
4	Data Testing - BAB III	Memindahkan penjelasan teori GCV ke Bab 2, sehingga Bab 3 hanya berfokus pada metodologi pengujian.
5	Kalender Tanam - BAB IV	Ditambahkan subbab Analisis Kalender Tanam dari Hasil Peramalan Menggunakan Holt-Winters dan LSTM.
6	Keseragaman Penulisan dan Penelitian Terkait	Penelitian terkait diperbarui, istilah diseragamkan (time-series, dataset, preprocessing, dsb.)

Flowchart Anggota 1



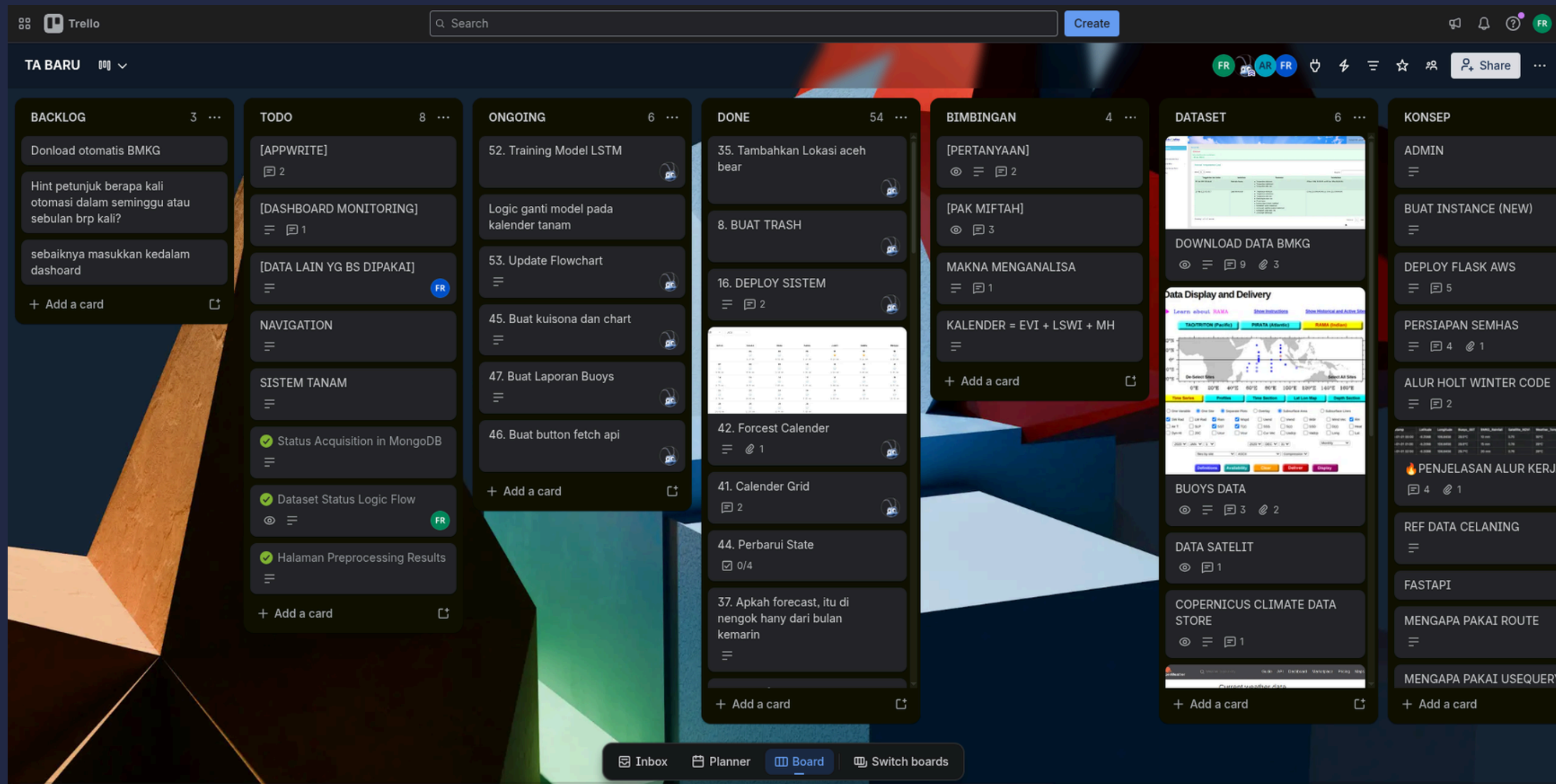
2. HASIL DAN PEMBAHASAN



Tabel Pembagian Sprint

Sprint	Fokus Pengembangan	Output Utama
1	Analisis kebutuhan	<i>Product backlog</i>
2	Perancangan dan implementasi <i>frontend</i>	<i>Prototype, UI dashboard</i>
3	Implementasi <i>backend</i> dan RESTful API	<i>Preprocessing endpoint</i>
4	Implementasi <i>smoothing</i>	Otomasi sistem
5	Pengujian dan <i>deployment</i>	<i>System live</i>

Implementasi Agile Development



Perbedaan Implementasi Smoothing pada BMKG dan NASA POWER

BMKG

```
Raw Data
|
Validasi
|
Imputasi
|
EWMA Lokal
(hanya hasil imputasi)
|
Dataset Final
```

Korektif dan Lokal

NASA POWER

```
Raw Data
|
Validasi
|
Imputasi
|
Smoothing Global
(seluruh time-series)
|
GCV + Trend Preservation
|
Dataset Final
```

Aktif dan Menyeluruh

Preprocessing BMKG

```
preprocessing_bmkg.py Python
1 # parameter_configs (line ~870)
2 "parameter_configs": {
3     "TAVG": {
4         "forecasting_critical": True,
5         "apply_smoothing_on_imputed": True, # ← aktifkan smoothing
6         "smoothing_alpha": 0.15,
7         "reason": "smooth only imputed points"
8     },
9     "RH_AVG": {
10        "forecasting_critical": True,
11        "apply_smoothing_on_imputed": True, # ← aktifkan smoothing
12        "smoothing_alpha": 0.15,
13        "reason": "smooth only imputed points"
14    },
15    "RR": {
16        "forecasting_critical": True,
17        "apply_smoothing_on_imputed": False, # ← tidak di-smooth
18    },
19 }
```

```
preprocessing_bmkg.py Python
1 # _apply_preprocessing (line ~992)
2 for param, flag_col in [("TAVG", "TAVG_imputed"),
3                          ("RH_AVG", "RH_AVG_imputed")]:
4
5     alpha = float(param_cfg[param].get("smoothing_alpha", 0.15))
6
7     # Mask: hanya titik yang diimputasi & tidak NaN
8     mask = (processed_df[flag_col].fillna(False)
9            & processed_df[param].notna())
10
11     if mask.any():
12         smoothed = processed_df[param].ewm(
13             alpha=alpha, adjust=True
14         ).mean()
15         # Tulis kembali HANYA pada titik imputasi
16         processed_df.loc[mask, param] = smoothed.loc[mask]
```

BMKG bersifat lokal dan pasif — hanya memperbaiki artefak sambungan imputasi, tanpa mengubah struktur data asli.
Sesuai dengan filosofi “**menjaga integritas observasi lapangan.**”

Preprocessing NASA POWER

```
preprocessing_nasa.py Python
1 # Dispatcher smoothing
2 for param in params:
3     method = param_config.get("smoothing_method", "exponential")
4
5     if method == "exponential":
6         df[param] = self._smooth_exponential(df[param], param)
7     elif method == "adaptive_exponential":
8         df[param] = self._smooth_exponential_adaptive(df[param], param)
9     elif method == "moving_average":
10         df[param] = self._smooth_moving_average(df[param])
11     elif method is None:
12         pass
```

```
preprocessing_nasa.py Python
1 # Adaptive exponential smoothing
2 volatility = series.rolling(window=30, min_periods=1).std()
3 volatility_norm = normalize_by_percentile(volatility)
4
5 alphas = alpha_min + volatility_norm * (alpha_max - alpha_min)
6
7 for i in range(1, len(series)):
8     smoothed[i] = (
9         alphas[i] * series.iloc[i]
10        + (1 - alphas[i]) * smoothed[i-1]
11    )
```

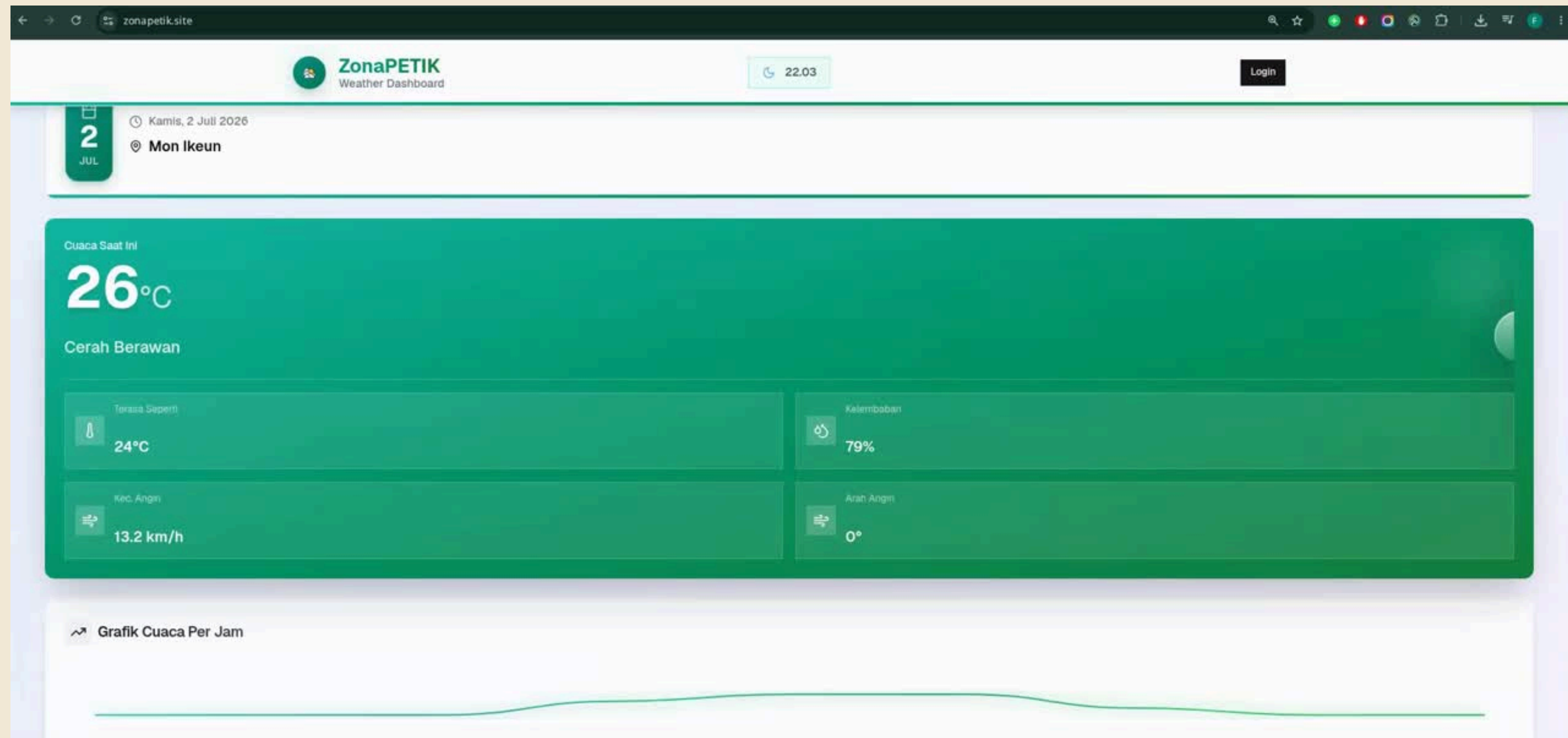
NASA bersifat global dan adaptif, sesuai dengan filosofi “**mengurangi noise seluruh dataset untuk kepentingan forecasting**”.

Dataset	Parameter	GCV	Status	Trend Preservation	Catatan
BMKG	RR	1.35	Excellent	90.12%	Varians tinggi - GCV Rendah
NASA	RH2M	3.07	Poor	82.25%	Varians rendah - GCV Tinggi

Hasil Perbandingan

- GCV mengukur error relatif terhadap varians data, bukan error absolut.
- Semakin tinggi varians data → semakin kecil GCV (dengan error yang sama).
- Trend preservation keduanya tetap > 80%, artinya smoothing masih mempertahankan pola jangka panjang.

Implementasi Pada Sistem Website



1. Halaman Dashboard

Tambah Dataset

Unggah file CSV/JSON/XLSX atau ambil data dari NASA POWER API.

Upload File

NASA POWER API

☐ Upload Multi-File XLSX (untuk menggabungkan beberapa file)

Nama Dataset *

Dataset BMKG Aceh Besar

Sumber

Data BMKG (<https://dataonline.bmkg.go.id/>)

Status

Raw

Raw

Cleaned

Validated

Upload File

Klik atau seret file .csv atau .json atau .xlsx ke sini

Maksimum 50MB

Batal

Simpan

Tambah Dataset

Unggah file CSV/JSON/XLSX atau ambil data dari NASA POWER API.

Upload File

NASA POWER API

Nama Dataset*

NASA Aceh Besar Kec. Indrapuri 2005

Deskripsi (Opsional)

Deskripsi singkat tentang dataset ini

Lokasi*

Kabupaten

Pilih Kabupaten

Kecamatan

Pilih Kecamatan

Latitude

-

Longitude

-

Tanggal Mulai *

Pilih tanggal mulai

Tanggal Akhir *

Pilih tanggal akhir

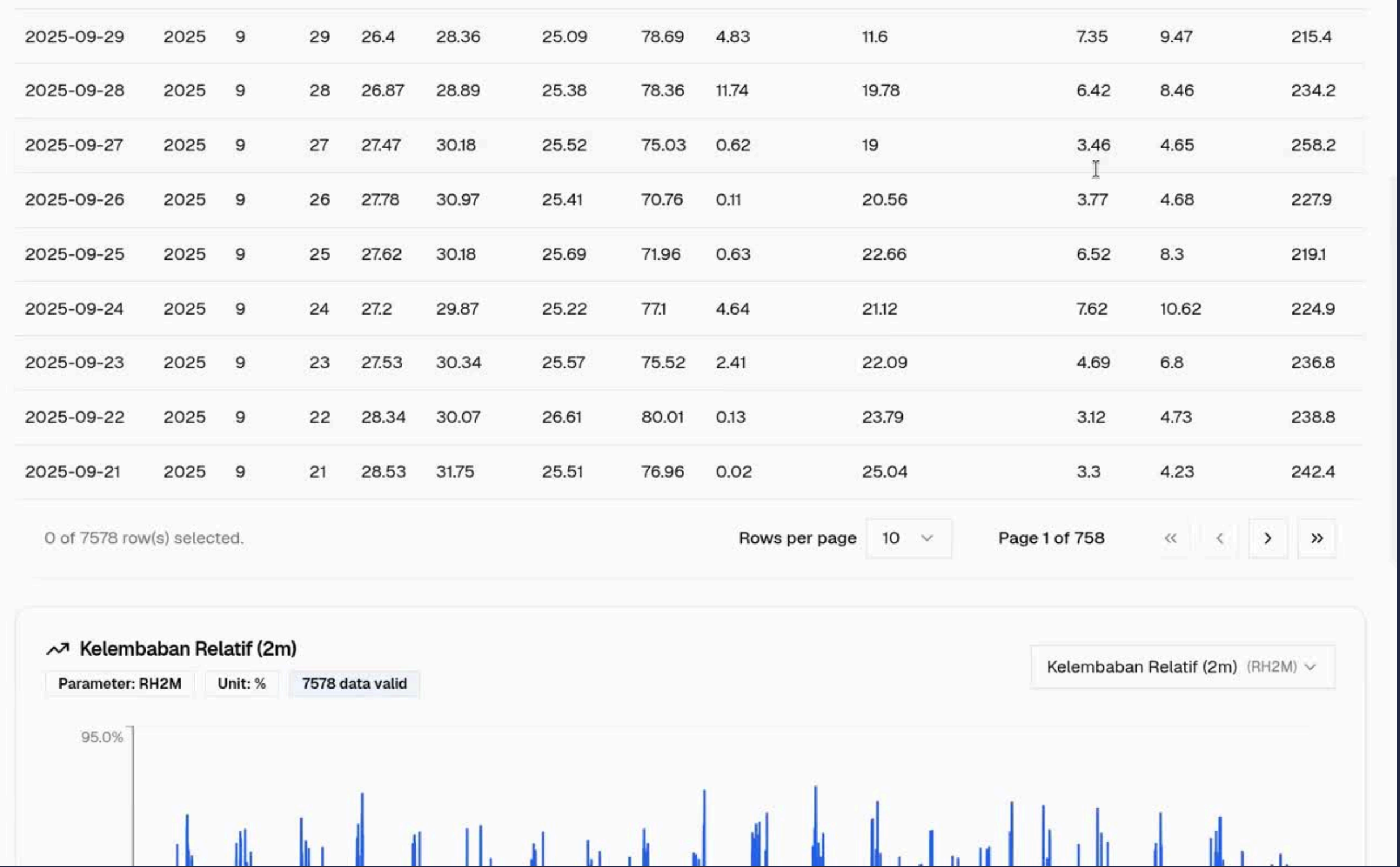
Pilih tanggal mulai dan akhir untuk menentukan periode

Preview

Batal

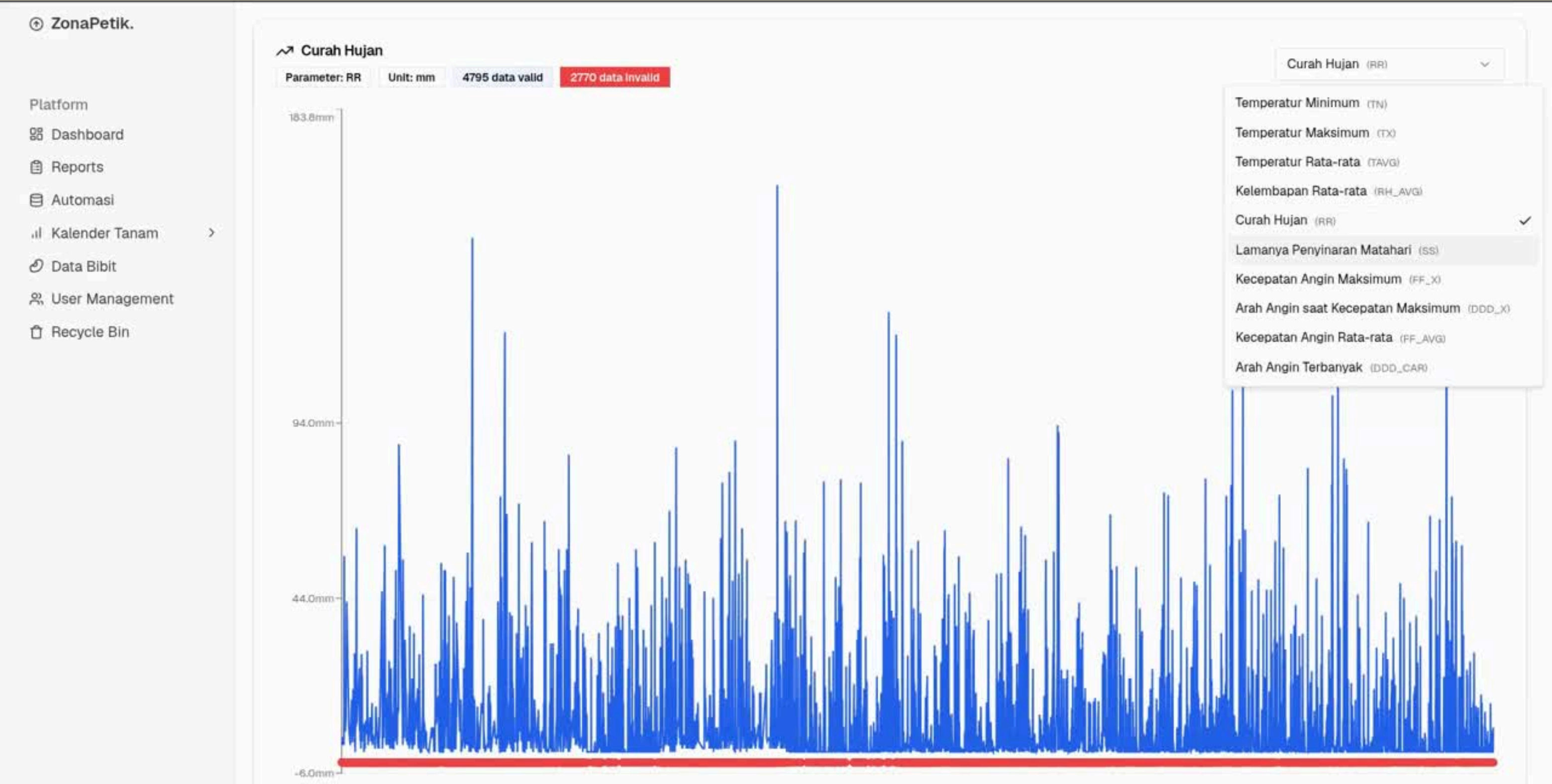
Simpan

2. Tambah Dataset



3.Refresh Dataset (NASA)

4.Detail Dataset



5. Preprocessing & Smoothing

```
tdmrc@compute-srg:~$ sudo systemctl restart flaskapp.service && journalctl -u flaskapp.service -f
[sudo: authenticate] Password:
Jul 02 11:21:15 compute-srg systemd[1]: flaskapp.service: Start request repeated too quickly.
Jul 02 11:21:15 compute-srg systemd[1]: flaskapp.service: Failed with result 'exit-code'.
Jul 02 11:21:15 compute-srg systemd[1]: Failed to start flaskapp.service - Gunicorn Flask App.
Jul 02 11:21:15 compute-srg systemd[1]: flaskapp.service: Consumed 5.699s CPU time over 129ms wall clock time, 150.3M memory peak.
Jul 02 21:15:17 compute-srg systemd[1]: Started flaskapp.service - Gunicorn Flask App.
Jul 02 21:15:17 compute-srg gunicorn[11714]: [2026-07-02 21:15:17 +0700] [11714] [INFO] Starting gunicorn 26.0.0
Jul 02 21:15:17 compute-srg gunicorn[11714]: [2026-07-02 21:15:17 +0700] [11714] [INFO] Listening at: http://0.0.0.0:5001 (11714)
Jul 02 21:15:17 compute-srg gunicorn[11714]: [2026-07-02 21:15:17 +0700] [11714] [INFO] Using worker: gevent
Jul 02 21:15:17 compute-srg gunicorn[11716]: [2026-07-02 21:15:17 +0700] [11716] [INFO] Booting worker with pid: 11716
Jul 02 21:15:17 compute-srg gunicorn[11714]: [2026-07-02 21:15:17 +0700] [11714] [INFO] Control socket listening at /home/tdmrc/.gunicorn/gunicorn.ctl
Jul 02 21:50:20 compute-srg gunicorn[11716]: ERROR:routes.scheduler_routes:Error fetching scheduler status: ac-m578nkv-shard-00-01.uuonyyb.nongodb.net:27017: [Errno -5] No address associated with hostname (conn
figured timeouts: connectTimeoutMS: 20000.0ms)
Jul 02 22:21:12 compute-srg gunicorn[11716]: INFO:app:BMKG Stream Request Headers: {'Host': 'backed-octagon-levitate.ngrok-free.dev', 'User-Agent': 'Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36', 'Accept': 'text/event-stream', 'Accept-Encoding': 'gzip, deflate, br, zstd', 'Accept-Language': 'en', 'Ngrok-Skip-Browser-Warning': '69420', 'Origin': 'https://zonapetik.site', 'Priority': 'u=1, i', 'Referer': 'https://zonapetik.site/', 'Sec-Ch-Ua': '"Chromium";v="148"', 'Google Chrome';v="148"', 'Not/A)Brand";v="99"', 'Sec-Ch-Ua-Mobile': '?0', 'Sec-Ch-Ua-Platform': 'Linux', 'Sec-Fetch-Dest': 'empty', 'Sec-Fetch-Mode': 'cors', 'Sec-Fetch-Site': 'cross-site', 'X-Forwarded-For': '2404:c0:9201:4a3a:d578:4e10:676b:2c7', 'X-Forwarded-Host': 'backed-octagon-levitate.ngrok-free.dev', 'X-Forwarded-Proto': 'https'}
Jul 02 22:21:12 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:PROGRESS: Starting BMKG preprocessing pipeline...
Jul 02 22:21:12 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:
Jul 02 22:21:12 compute-srg gunicorn[11716]: [1/7] Validating dataset...
Jul 02 22:21:12 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Validating 7565 records in Dataset BMKG Aceh Besar...
Jul 02 22:21:12 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Validation passed - 7565 records
Jul 02 22:21:12 compute-srg gunicorn[11716]: WARNING:preprocessing.bnkg.preprocessing_bnkg:Validation warnings (1):
Jul 02 22:21:12 compute-srg gunicorn[11716]: WARNING:preprocessing.bnkg.preprocessing_bnkg: - Field 'RR' has 70.0% missing values (high!)
Jul 02 22:21:12 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:
Jul 02 22:21:12 compute-srg gunicorn[11716]: [2/7] Loading data from MongoDB...
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Successfully loaded 7565 records from 'Dataset BMKG Aceh Besar'
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Loaded 7565 records
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:
Jul 02 22:21:13 compute-srg gunicorn[11716]: [3/7] Applying preprocessing pipeline...
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:PROGRESS: Starting preprocessing steps...
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Starting preprocessing steps
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:[1/6] Preparing temporal features...
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg: Date range: 2005-01-01 00:00:00 to 2025-09-30 00:00:00
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg: Temporal features added: Season, Month
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:[2/6] Replacing fill values...
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:FF_AVG ZERO-INFLATION DIAGNOSTIC (BEFORE FIX)
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Total records: 7565
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:FF_AVG = 0.0: 1738 (23.0%)
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:FF_AVG missing: 566 (7.5%)
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:FF_AVG > 0 (valid): 5261 (69.5%)
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Suspicious zeros (FF_X>0 but FF_AVG=0): 1576 (20.8%)
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Replaced fill values: {'RR': 1994, 'SS': 4}
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Detected 1576 suspicious FF_AVG zeros
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Sample suspicious zeros:
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:Date FF_X FF_AVG (orig) Action
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:2005-01-15 4.0 0.0 → Replace with NaN
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:2005-01-16 3.0 0.0 → Replace with NaN
Jul 02 22:21:13 compute-srg gunicorn[11716]: INFO:preprocessing.bnkg.preprocessing_bnkg:2005-01-18 3.0 0.0 → Replace with NaN
```

Preprocessing NASA POWER Data

NASA Dataset Report bab 4

Calculating model coverage analysis...

90%

Live Logs (180)

Auto-scrolling

8:30:42 PM INFO:preprocessing.nasa.preprocessing_nasa:Validating smoothing quality...

8:30:42 PM INFO:preprocessing.nasa.preprocessing_nasa:Parameter T2M: GCV=0.0614, Trend=83.48%, Quality=GOOD

8:30:42 PM INFO:preprocessing.nasa.preprocessing_nasa:Parameter T2M_MAX: GCV=0.2475, Trend=82.57%, Quality=GOOD

8:30:42 PM INFO:preprocessing.nasa.preprocessing_nasa:Parameter T2M_MIN: GCV=0.0511, Trend=82.53%, Quality=GOOD

8:30:42 PM INFO:preprocessing.nasa.preprocessing_nasa:Parameter RH2M: GCV=3.0746, Trend=82.25%, Quality=POOR

8:30:42 PM INFO:preprocessing.nasa.preprocessing_nasa:Smoothing validation completed for 4 parameters

8:30:42 PM [90%] Calculating model coverage analysis...

8:30:42 PM INFO:preprocessing.nasa.preprocessing_nasa:Calculating model coverage analysis...

8:32:24 PM WARNING:preprocessing.nasa.preprocessing_nasa:Parameter PRECTOTCORR: Very weak seasonality (strength=0.115) - Holt-Winters may be inappropriate

8:32:25 PM INFO:preprocessing.nasa.preprocessing_nasa:PRECTOTCORR: Halving outlier penalty for LSTM due to precipitation characteristics.

Stop & Close

ZonaPetik. Platform Dashboard Reports Automasi Kalender Tanam Data Bibit User Management Recycle Bin	Original Series: TEST CASE 2 NASA PREPROCESSING... 7,578 Records Cleaned on 18 Mei 2026 View Results	Original Series: bkmkg test stream preprocessing on deploy 7,565 Records Cleaned on 16 Mei 2026 View Results	Original Series: NEW NASA POWER ON tdmrc server 7,578 Records Cleaned on 16 Mei 2026 View Results
	Nasa power test preprocessing from vercel (Cleaned) Original Series: Nasa power test preprocessing from vercel 7,796 Records Cleaned on 12 Mei 2026 View Results	NASA Dataset Report Preprocessing Bab 4 (Cleaned) Original Series: NASA Dataset Report Preprocessing Bab 4 7,578 Records Cleaned on 11 Mei 2026 View Results	NASA updating data via host edited (Cleaned) Original Series: NASA updating data via host 7,795 Records Cleaned on 10 Mei 2026 View Results
	BMKG Dataset Report Preprocessing Bab 4 (Cleaned) Original Series: BMKG Dataset Report Preprocessing Bab 4 7,565 Records Cleaned on 09 Mei 2026 View Results	bkmkg preprocessing third run (Cleaned) Original Series: bkmkg preprocessing third run 7,573 Records Cleaned on 07 Mei 2026 View Results	Nasa test for second run (Cleaned) Original Series: Nasa test for second run 7,792 Records Cleaned on 07 Mei 2026 View Results
	Dataset Nasa 1 Update (Cleaned) Original Series: Dataset Nasa 1 Update 7,792 Records Cleaned on 07 Mei 2026 View Results	Dataset BMKG Preprocessing (Cleaned) Original Series: Dataset BMKG Preprocessing 7,573 Records Cleaned on 07 Mei 2026 View Results	Dataset Nasa 2 Preprocessing (Cleaned) Original Series: Dataset Nasa 2 Preprocessing 7,578 Records Cleaned on 07 Mei 2026 View Results

6.Preprocessing Report

⌕ ZonaPetik.

Platform

Dashboard

Reports

Automasi

Kalender Tanam

Data Bibit

User Management

Recycle Bin

Monitor execution logs, trigger manual processes, and configure recurring background tasks for your climate datasets.

Scheduler Status

● Active

▶ Run Scheduler Now

🕒 LAST RUN

Jun 29, 2026, 04:00 PM GMT+7

Duration: 7m 14s

Success • 3/3 Datasets

🕒 NEXT RUN

Jul 6, 2026, 04:00 PM GMT+7

(In 89h 32m)

📈 PERFORMANCE (30 DAYS)

Success Rate 100%

3 / 3 successful runs

Avg Duration: 7m 18s

Loading config...

Execution History

All Status ▾ ↺

DATE & TIME	DURATION	STATUS	TASKS	DETAILS
Jun 29, 04:00 PM	7m 14s	Success	3/3 updated	> View

7.Automation Config

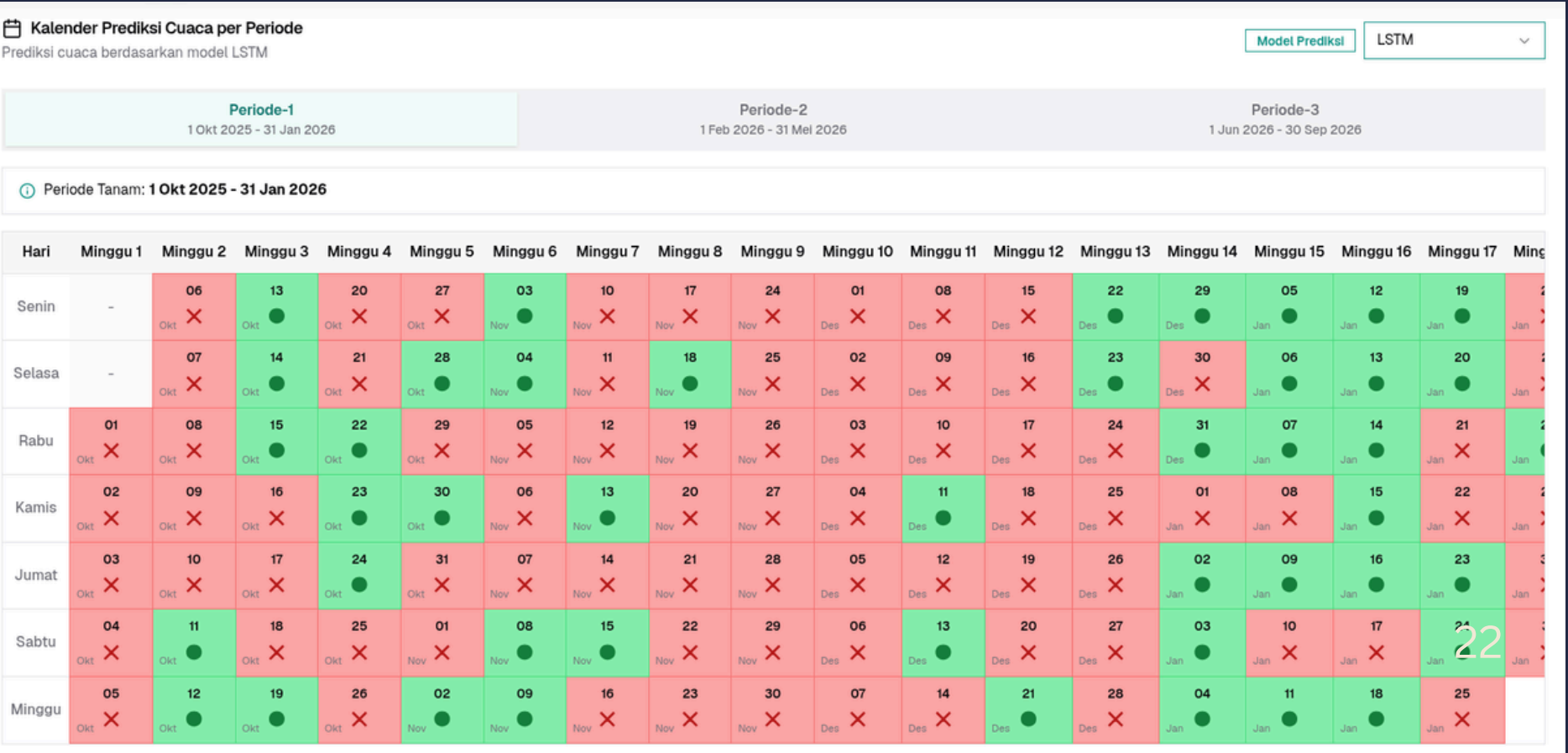
21

Kalender Tanam

LSTM

Holt-Winters

Metode	MAE	RMSE	MAPE (%)
Holt-Winters	5.65	13.71	60.83
LSTM	0.156	0.242	9.954



Kesimpulan

Smoothing berhasil meningkatkan kualitas data. BMKG: suhu & kelembaban excellent (trend >99,9%), RR 90,12%. NASA: suhu good (83,48%), kelembaban tetap pertahankan tren (82,25%).

Otomasi & API berhasil diimplementasikan (Next.js, Flask, MongoDB, cron). 25 endpoint REST API berfungsi, sistem terintegrasi secara terpadu.

Kalender tanam LSTM lebih adaptif dan presisi daripada Holt-Winters. LSTM memberikan margin keamanan panen lebih besar (59–62 hari vs 4–12 hari) dan kategori kesesuaian lebih rinci.

Saran

Tingkatkan imputasi & smoothing pada RR (BMKG) dan RH2M (NASA) dengan metode alternatif

(wavelet denoising, ensemble smoothing, atau data satelit GPM).

Perluas pengujian dengan load testing & stress testing untuk evaluasi performa API, scheduler, dan database pada skenario beban besar.